Wireless Sensor Network

1. Fundamentals of Wireless Communication Technology

Source → Input Transducer → Transmitter → Communication channel + Noise → Receiver → Output Transducer → Destination

Communication System

* **What is Wireless Communication?**

↳ Communication without wires using electromagnetic waves.

* **Basic Components:**
  - Transmitter → sends signals
  - Channel → air (wireless medium)
  - Receiver → receives signal

* **Types:**
  - Radio (FM/AM)
  - Wifi
  - Bluetooth
  - Satellite

2. Electromagnetic Spectrum :-

↳ Definition :- Range of all electromagnetic waves arranged by frequency.

* **Order :-**

Radio → Microwave → Infrared → Visible → UV → X-ray → Gamma.

* **Used in WSN:**

- Mostly Radio Waves (low frequency)

3. Radio Propagation Mechanisms

* How signals travel:
  1. Reflection → bounce from surfaces.
  2. Differection → bend around obstacles.
  3. Scattering → Spread due to small objects.

- Problem:

⇒ Signal may weaken = fading

4. Characteristics of Wireless Channel

- Key properties:
  - Path loss → Signal weaken over distance
  - fading → signal fluctuation.
  - Interface → Other Signals disturb
  - Noise → Unwanted signals.

$\boxed{\text{Signal} \downarrow \text{ when distance } \uparrow}$

5. MANETs & WSNs

- MANET (mobile Ad-hoc Network)
  - No fixed infrastructure
  - Nodes move freely
  - Example: mobile-to-mobile network.
- WSN (Wireless Sensor Network)
  - Many small sensor nodes
  - Collect data (temperature, humidity, etc.)
  - Send to base Station.

#. WSN Vs MANET

	WSN	MANET
Goal	Detection/estimation of some events of interest	Simply communications.
Communication Pattern	Specialized to: Many-to-one One-to-many Local communications.	Typically support routing b/w any pair of nodes.
energy/ resources constrained	More	Less
Mobility	Mostly not mobile	Mostly mobile
Cooperation among nodes	More cooperative, exhibit trust relationships	Less cooperative
Security mechanism	Authentication and routing based on public key cryptography is too expensive.	Both public key or asymmetric cryptography can be applied
Routing	Distance vector and Source routing protocols are generally too expensive	Support different types of routing protocols.

6. Concepts & Architecture of WSN

- Sensor Node Parts:
  - Sensor
  - Microcontroller
  - Battery
  - Transceiver

- Architecture Types:
  1. Flat architecture
  2. Clustered architecture

- Flow:

Sensor → Node → Sink/Base Station → User

7. Applications of WSN

* Real Uses:
  - Agriculture (soil monitoring)
  - Healthcare (patient monitoring)
  - forest fire detection
  - Military Surveillance
  - Smart cities

8. Design Challenges in WSN

- Major problems:
  → Energy constraint
  → Limited bandwidth
  → Security issues
  → Scalability
  → fault tolerance

→ "Energy efficiency is the biggest challenge in WSN"

Module-2: MAC Protocols

1. Issues in Designing a MAC Protocol.

- What is MAC ?
  ↳ medium Access Control.

→ Decides who can use the channel at what time.

#. Major Problems:

1. Hidden Terminal Problem
   - Two nodes can't see each other
   - Both send → Collision occurs.

   Ex:- A and C send to B (but A & C don't know each other)

2. Exposed Terminal Problem
   - Node stops sending unnecessarily
   - even when transmission is safe.

3. Collision
   - Multiple nodes transmit at same time
   - Data lost.

4. Energy efficiency
   - Battery is limited (WSN)
   - Idle listening wastes power

→ "MAC design must handle hidden terminal, exposed terminal, Collision and energy efficiency"

2. Classification of MAC Protocols.

- **Multiple Access Protocols**
  - **Random Access Protocol**
    - Aloha
    - CSMA
    - CSMA/CD
    - CSMA/CA
  - **Controlled Access Protocol**
    - Reservation
    - Polling
    - Token Passing
  - **Channelization Protocol**
    - FDMA
    - TDMA
    - CDMA

Types :

1. Contention - Based
   - Nodes compete
   - Example : CSMA
2. Schedule - Based
   - fixed time slots
   - example : TDMA
3. Hybrid
   - Combination of both.

3. Contention - Based Protocols.

↳ "Listen before transmit"

Ex:- CSMA / CA (Carrier Sense Multiple Access with Collision Avoidance)

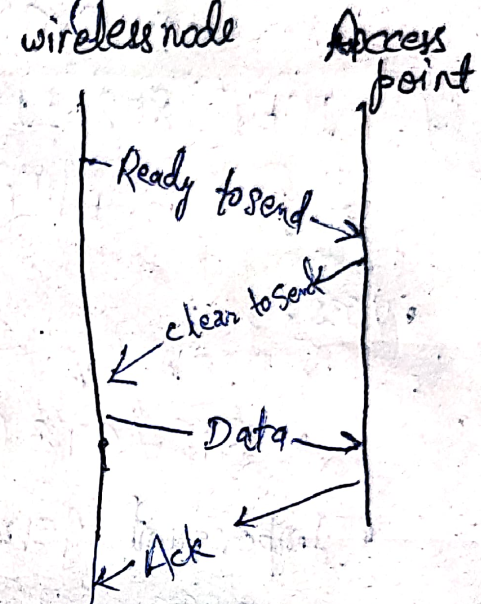
* Working :

1. Check channel
2. If free → send
3. If busy → wait (backoff)

• Advantage:
Simple.
• Disadvantage:
collision still possible.

4. Contention - Based with Reservation.

Idea: Reserve channel before sending



• RTS/CTS Mechanism:
• RTS = Request to Send
• CTS = clear to Send.

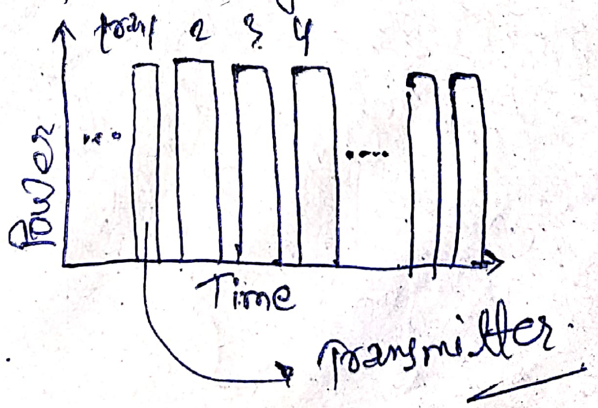
• Flow:
Sender → RTS → Receiver
Receiver → CTS → Sender
Then → Data send

• Advantages:
Reduce Collision

5. Contention - Based with Scheduling.

Idea:- Each node get time slot.

Time Division multiple Access (TDMA)
• Allocation of time slot in TDMA



Ex:- TDMA

Working:-
• Time divided into slots
• Each node sends in its slot.

Advantage:
• No Collision
• energy efficient

Disadvantage:- Synchronization required.

6. Multi-Channel MAC

↳ Use multiple channels instead of one.

- Advantages:
  - less collision
  - High throughput
- Problem:
  - channel coordination needed.

7. IEEE 802.11 (WiFi - MAC)

↳ Used in WiFi networks

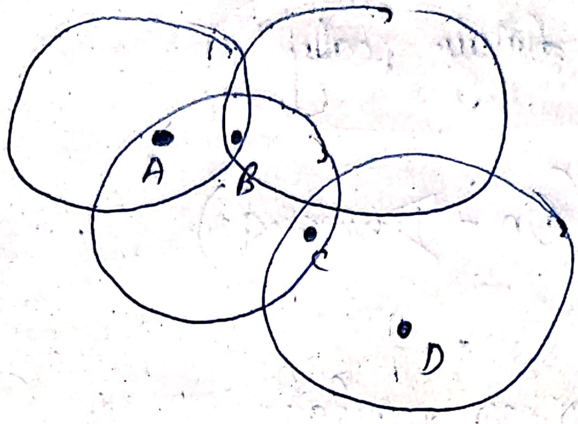
- Uses:
  - CSMA/CA
  - RTS/CTS (optional)
- Important Terms:
  - SIFS → Short Inter frame Space
  - DIFS → Distributed Inter frame Space.

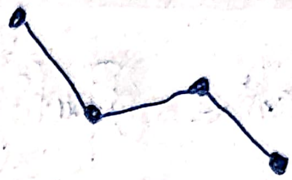
Module-3: Routing & Transport Layer

1. Issues in Designing Routing & Transport Protocols.

Mobile Ad Hoc Networks.
(MANET)

- Mobility causes route changes





Only one possible route: 3 hops
A-B-C-D

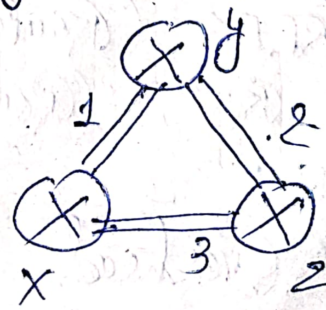
- Why difficult?

↳ Ad-hoc networks are dynamic (nodes move)

- **Major Issues:**
  - Dynamic Topology → nodes keep moving
  - Link failure → routes break frequently
  - Limited Bandwidth
  - Energy Constraint
  - Routing Overhead → too many control packets.

2. Proactive Routing

→ Idea:
→ Maintain routes all the time.



	x	y	z
x	0	1	3
y	1	0	2
z	3	2	0

Ex:- DSDV (Destination-Sequenced Distance Vector)

✓ Advantages :
→ Low delay (route already known)

✗ Disadvantages :
→ High overhead (updates always)

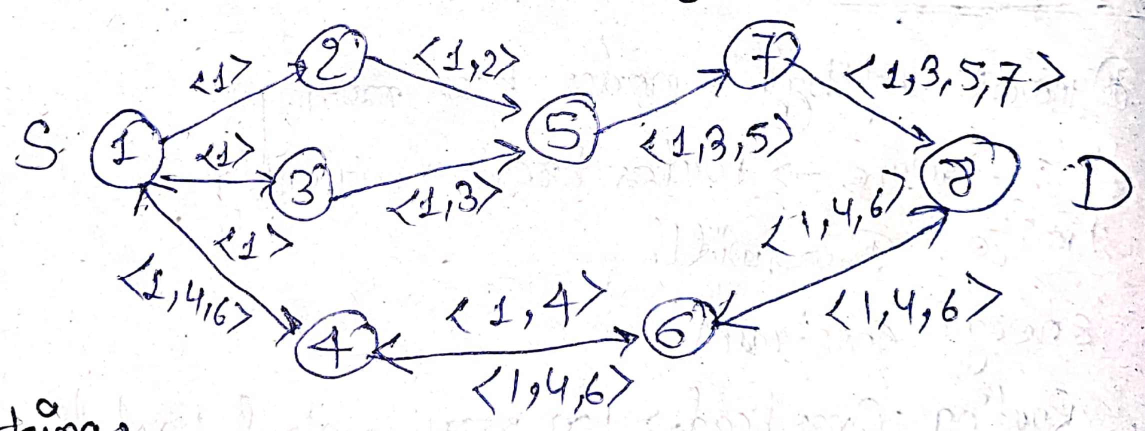
"Proactive = route always ready"

3. Reactive Routing (On-Demand)

Idea :
→ Create route only when needed

Ex:- • AODV
• DSR

• Dynamic Source Routing (DSR)



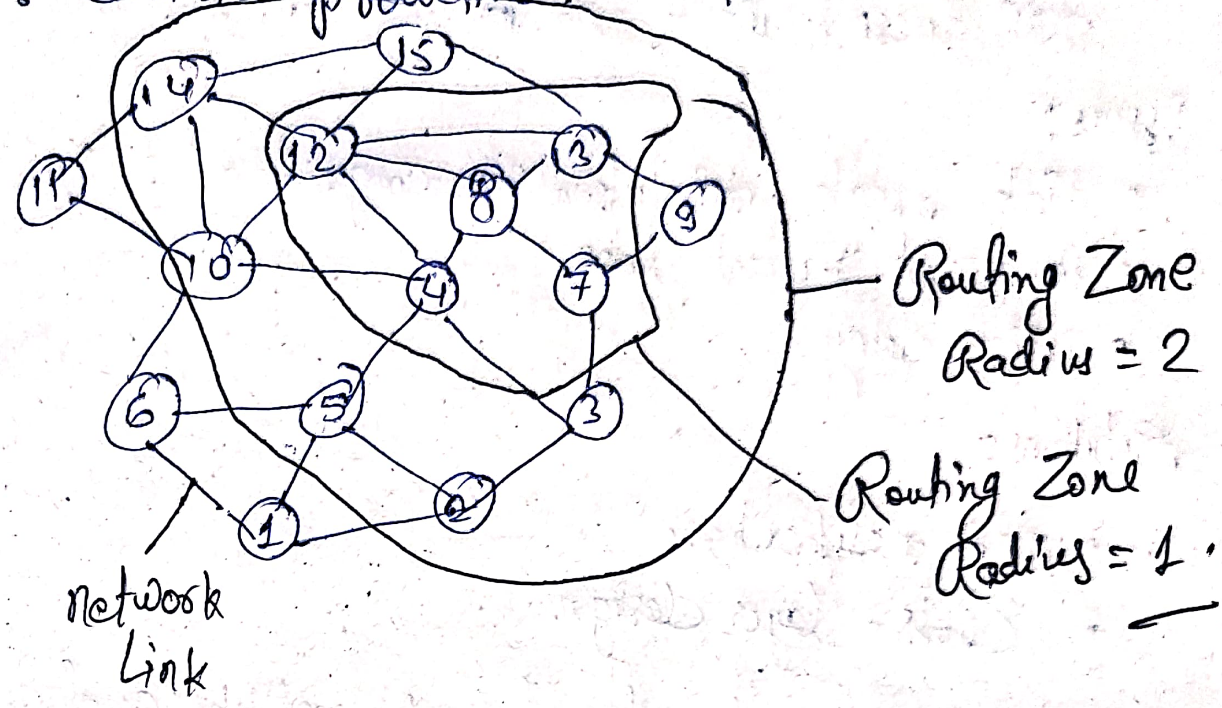
• Working :
1. Send RREQ (Route Request)
2. Receive RREP (Route Reply)

• Advantages :
• Less Overhead

• Disadvantages :
• Delay in route discovery.

4. Hybrid Routing

Idea :- Combine proactive + reactive.



Ex:- ZRP (Zone Routing Protocol)

- Advantage:
  - Balanced performance.

5. Classification of Transport Layer Sol^n.

* Types:

1. Reliable (TCP-based)
   - Ensures delivery
2. Unreliable (UDP-based)
   - faster but no guarantee.

- Problem in Ad-hoc:

TCP assumes loss = congestion.

But in wireless → loss can be due to link failure.

6. TCP over Ad-hoc Networks.

- Problem:
  TCP doesn't work well in ad-hoc.

Issues:

- Link break mistaken as congestion
- Frequent packet loss
- High delay

Solutions:

- TCP modifications
- Cross-layer design

"TCP performance degrades due to mobility and link failures".

Q.2.(a.) What is a Blockchain ?

↳ **Definition :** Blockchain is a distributed, decentralized digital ledger that stores data in blocks, linked using cryptography.

Key Points :

- Data stored in blocks
- Blocks connected like a chain
- No central authority (decentralized)
- Data is immutable (cannot be change)

Ex:- Bitcoin is the first real blockchain application.

(b.) What are the components of a blockchain architecture ?

↳ There are mainly 5 components :

1. Node
   - A computer connected to the blockchain network
   - Stores a copy of the blockchain.

2. Block
   - Contains :
     - Data (transactions)
     - Hash (Unique ID)
     - Previous block hash

3. Chain
   - Blocks are connected in sequence
   - Forms a continuous chain.

4. Miner
   - Verifies transactions
   - Adds new blocks to the chain.

5. Consensus Mechanism
   - Rules to agree on valid transactions.

Q.3.) (a.) How does a Cryptocurrency work?

↳ A cryptocurrency is a digital money system that works on blockchain.

Step - by - step :

1. User sends money
2. Transaction is broadcast to network
3. Node verify it
4. Miner adds it to a block
5. Block is added to blockchain.

Important features :

- No bank needed
- Secure using cryptography
- Transparent

(b.) What is difference between Bitcoin and Blockchain?

↳ feature	Bitcoin	Blockchain
• Meaning	Digital currency	Technology
• Purpose	Used for payment	Stores data
• Type	Application	Platform
• Example	Bitcoin	Used in many system
• What is it?	A cryptocurrency	A distributed Database.

Simple Understanding :
- Blockchain = Road
- Bitcoin = car running on that Road.

• Main Aim	To simplify and increase the speed of transactn without much of government restrictions.	To provide a low cost, safe & secure environment for peer to peer transactn.

5. What is Consensus Mechanism in Blockchain?

↳ A concensus mechanism is a method used to agree on valid transactions in a blockchain network.

Why needed?

- No central authority
- So all nodes must agree

Types of Consensus:

1. Proof of Work (PoW)
   - Used in Bitcoin
   - miners solve complex problems
   - High energy consumption
2. Proof of stake (POS)
   - Based on coin ownership
   - less energy usage
3. Proof of Burn
   - Coin are destroyed to gain power
4. Proof of Elapsed Time
   - Random selection of nodes

- Advantages:
  - security
  - Trust
  - No fraud